

Motor Datasheet: WxF-50x27SL

1820 mk1

Operating Point: Continuous | Temperature: 20 °C

Geometry (active elements only)

Parameter	Value	Unit
Machine Outer Diameter	48.0	mm
Machine Inner Diameter	26.4	mm
Machine Thickness / Width	26.3	mm
Air Gap	0.50	mm
Fill Factor (input)	0.60	-
Slot Fill Factor (calculated)	Not calculated	-
Slot Width (tooth to tooth)	3.141	mm
Available Coil Space	1.570	mm
Assumed Coil Width (simulation)	1.370	mm

Mass Properties (active elements only)

Parameter	Value	Unit
Machine Mass	181.3	g
Rotor Mass	49.8	g
Stator Mass	131.6	g
Rotor Structure Included	No	-
Rotor Moment of Inertia (active elements only)	182.0	g·cm²

Electrical Properties

Parameter	Value	Unit
Supply Current (RMS)	3.962	A
Supply Current (Peak)	5.603	A
Id (Peak)	-0.000	A
Iq (Peak)	5.603	A
Phase Advance Angle	0.0	°elec
Phase Resistance	446.247	mOhm
D-Axis Inductance (Ld)	1144.311	μH
Q-Axis Inductance (Lq)	1034.185	μH
Terminal Inductance (Ltt = Ld + Lq)	2178.495	μH
PM Flux Linkage	8.126	mWb
Frequency at operating point (electrical)	416.667	Hz
Winding Configuration	wye (star)	-
DC Supply Voltage	48.000	V
Power Factor	0.855	-

Performance

Parameter	Value	Unit
Torque (Continuous)	702.2	mNm
Speed	2500.0	rpm
BackEMF Constant (Ke, phase peak)	8.5	mV/rpm
Motor Constant (Km, only resistive losses)	153.2	mNm/√W
Specific Motor Constant (Km / mass)	844.6	mNm/√W/kg
Torque Constant (Kt, peak, no-load)	123.2	mNm/A
Torque Ripple	2.0	%

Saturation	0.0	%
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Losses

Parameter	Value	Unit
Total Losses	22.9	W
Resistive Losses	21.0	W
Iron Core Losses	1.6	W
Magnet Eddy Current Losses	0.2	W
Rotor Structure Eddy Current Losses	0.0	W
Winding Eddy Current Losses	0.0	W

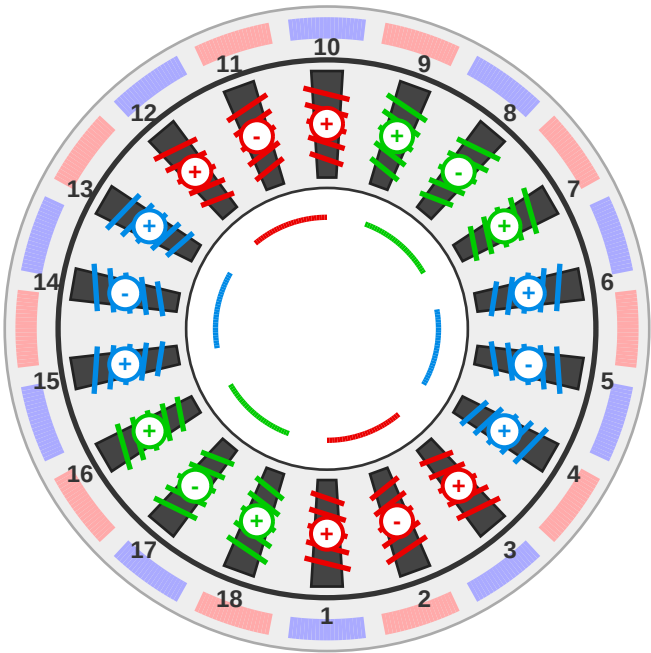
Configuration

Parameter	Value	Unit
Number of Stators	1	-
Number of Rotors	1	-
Number of Slots	18	-
Pole Pairs	10	-
Number of Parallel Paths	1	-
Number of Turns per Tooth (designed incl. parallel paths)	43.0	-
Number of Phase Turns (all phases in series)	258.0	-

Winding Configuration

Schema: AaABbBCcCAaABbBCcC

Concentrated Winding | Balanced | Cogging: 180/rev | Ripple: 180/rev



Materials

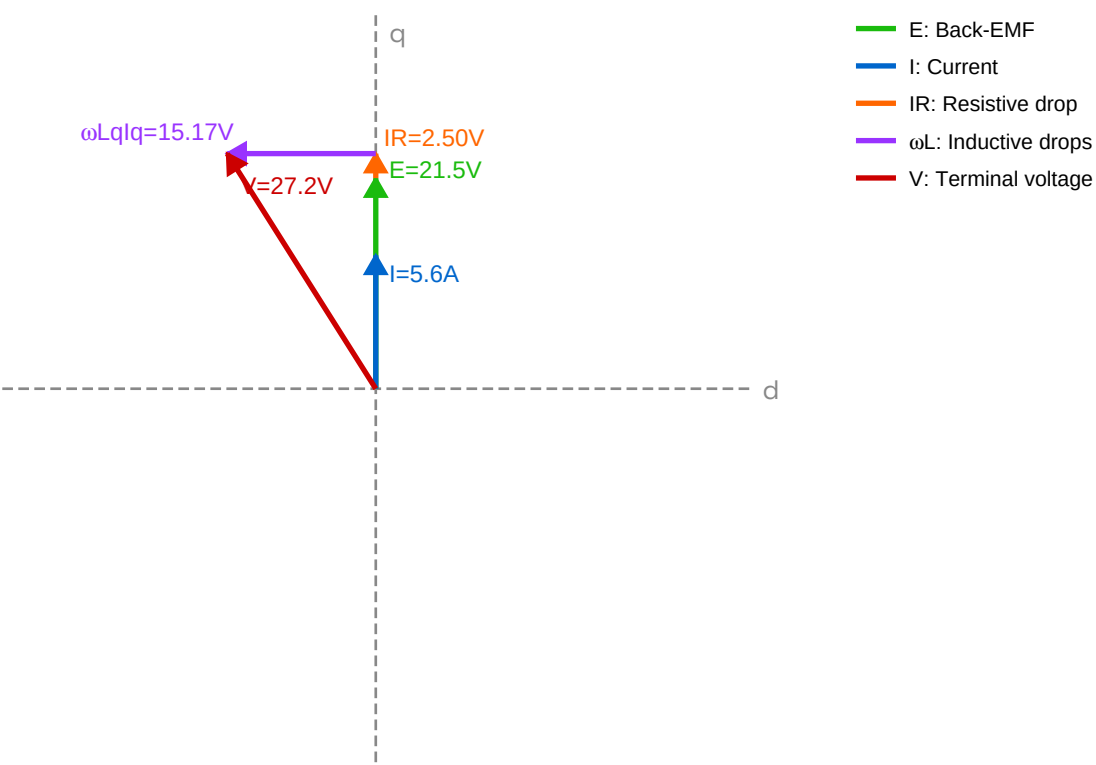
Component	Material	
Stator Core	N030-Cogent	-
Magnets	BMN-52SH (GBD)	-
Rotor Backiron	AISI 1010 Annealed	-
Coil	Copper	-

Voltage Breakdown at Operating Point (with integer turns)

Parameter	Value	Unit
Back-EMF Voltage (Phase, Peak)	21.502	V
Resistive Drop (Phase, Peak)	2.500	V
Inductive Drop Ld (Peak)	0.000	V
Inductive Drop Lq (Peak)	15.171	V

Inductive Drop Total (Peak)	15.171	V
Terminal Voltage (Phase, Peak)	27.236	V
Back-EMF Voltage (Line-to-Line, Peak)	37.242	V
Resistive Drop (Line-to-Line, Peak)	4.331	V
Inductive Drop (Line-to-Line, Peak)	26.277	V
Terminal Voltage (Line-to-Line, Peak)	47.173	V

Phasor Diagram Phase Values NOT Line-to-Line (d-q reference frame)



Mechanical Properties

Parameter	Value	Unit
Airgap Axial Force (Average)	211.39	N
Magnet Axial Force Positive Maximum	0.00	N
Magnet Axial Force Positive Minimum	0.00	N
Magnet Axial Force Negative Maximum	-54.55	N
Magnet Axial Force Negative Minimum	-44.00	N

Glossary

- Peak — Refers to 0-to-peak amplitude (NOT peak-to-peak). For sinusoidal waveforms: $\text{Peak} = \text{RMS} \times \sqrt{2}$.
- RMS — Root Mean Square, the equivalent DC value for power calculations.
- Line-to-Line (L-L) — Voltage measured between two phases. For wye connection: $V_{LL} = V_{\text{phase}} \times \sqrt{3}$.